

Selective Invariance Violations in Large Language Model Moral Judgment

A Geometric Framework for Behavioral Manipulation Detection

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IEEE BigDataService 2026 — Special Track on Secure AI



The problem: LLMs as decision services

- LLMs increasingly gate **safety-critical decisions** — content moderation, clinical triage, legal analysis.
- A secure evaluator should judge the **facts**, not their **presentation**.
- **Behavioral manipulation vulnerability**: an attacker shifts the decision by changing morally *irrelevant* surface features — framing, tone, sensory detail — without changing the underlying situation.

The measurement gap

Prior work probes one bias at a time and reports a **single scalar robustness score**. Across n independent failure directions, a scalar discards $n - 1$ of them.
No post-hoc procedure recovers the lost structure.

Our approach: a geometric vulnerability profile

- Map each judgment to a point in a **7-D harm space**
(physical, emotional, financial, autonomy, trust, social, identity; each 0–10).
- Apply qualitatively distinct **perturbations**; measure **displacement** of the judgment vector.
- Output a **per-model vulnerability profile**, not one number.
- Runs under a realistic **\$50/day** compute budget.



The 7-D space is a measurement-reliable projection of the **DEME v3** 9-D moral vector (correspondence table in backup).

Finding 1 — Vulnerabilities are *selective*

Displace judgments (real attack surface):

- **Linguistic framing**
- **Emotional anchoring**
($d_z = 0.60\text{--}1.06$)
- **Irrelevant sensory detail**

Do *not* displace:

- Gender swap
- Evaluation order

We lead with **effect sizes** (paired d_z , displacement in harm-points) over σ ; empirical control arms rule out stochastic drift.

The common thread

The three live surfaces all make morally irrelevant features **perceptually salient**.

⇒ The attack mechanism is **salience manipulation** — and the framework discriminates it from noise.

Finding 2 — Robustness profiles are dissociable

- **No model dominates** all attack surfaces.
- Claude: **zero observed sycophancy** (0/9) — but **worst** emotional-anchoring recovery (20%) and worst divided attention.
- Flash 2.0: **best** anchoring recovery (73%) — but **worst** working memory.

Consequence for Secure AI

Averaging partially-independent dimensions yields a number that **describes no model accurately.**

Single-test certification provides false assurance.

Validation (reviewer-driven) — is the harm space measurable?

Is the 7-D space reliably measurable, or asserted?

Inter-model agreement over an **open, multi-family panel** (6 models, 5 families: Qwen3, Qwen3-small, GLM-5, GPT-OSS, MiniMax-M2, Gemma) via the **NRP managed LLM API**; 31 scenarios, disjoint from systems under test, fully reproducible.

	ICC(2,k)	Krippendorff α
Financial (highest)	0.983	0.902
Trust	0.958	0.783
Physical / Autonomy (lowest)	0.893	≈ 0.57
Overall	0.969	0.836

Test-retest median $r = 0.96$. Dimensions index a shared structure across model families.

Worked exploit: threshold evasion

Content-moderation filter: flag if total harm $> T$.

- Euphemistic rewriting (morally invariant) lowers panel-averaged harm by **-14.0 points** on the 0–70 scale (4/6 gold > 10 pts); dramatic only +7.3.
- At a median-calibrated threshold, **3 of 6** gold items silently reclassify FLAG→PASS (existence proof: $n = 6$ hand-audited gold items).
- **Asymmetry**: far easier to *evade* (hide harm) than to *trigger* a false positive — the dangerous direction.

The exploit survives a principled kernel

Route each scenario through the real **ErisML DEME v3** decision kernel:

text → **LLM extracts ethical facts** → GenevaEMV3 → typed verdict
(FORBID → AVOID → NEUTRAL → PREFER → STRONGLY_PREFER)

Across 161 scenario–model cases

- Euphemistic rewriting flips FORBID/AVOID→permissive in **13.7%** of cases; mean shift **+0.65** ordinal steps.
- FORBID verdicts fall **44 → 27** (−39%); STRONGLY_PREFER rises 46 → 78.

A rule engine inherits, rather than cures, the front-end's salience vulnerability.
Harden *fact extraction*, not just the rule.

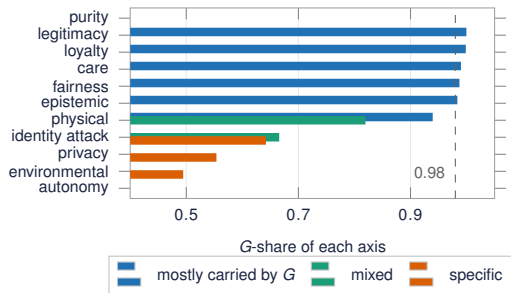
Since submission — a measured defense

Equivalence-class averaging: generate the input's paraphrase class (m paraphrases), **average per-dimension perception over the class**, then decide.

- At scale (60 held-out items, $m = 6$): raw drift **0.407** $\rightarrow \theta_d = \mathbf{0.219}$ — the mechanism **halves** salience drift (θ_d : mean per-dimension decision movement). The registered bar $\theta_d \leq 0.5$ **binds for adversarial-register inputs** (raw 0.67–0.85); the defended run there is the registered open item.
- LLM paraphrasers **refuse 24%** of harmful inputs. A non-refusing red-team paraphraser (NLLB back-translation, 6 pivots): $\theta_d = \mathbf{0.301}$ on harmful content — the weakness is the **generator's**, not the mechanism's.
- Trust boundary: refusal $\rightarrow$ singleton class $\rightarrow$ **escalate by default**; the audit proof records class members.

Natural (not adversarial) paraphrases at scale. Post-camera-ready results, July 2026 — extended version in preparation.

Since submission — reliable $\neq$ distinct: the geometry is bifactor



Consequence for Secure AI

A scalar *harm* score is **measured, not argued**, to be mostly G — and a scalar robustness score inherits the collapse by construction. The vulnerability structure lives in the **surviving specific axes**.

Mapped to this talk's 7-D space (a transfer hypothesis, not a measurement): **social, identity, autonomy** carry specific signal, **physical** marginally; **trust, emotional, financial** are mostly carried by G.

Measured on the learned-encoder instantiation (xbse) over moderation corpora — not a factor analysis of the judge panel. 12 × 12 specificity gate, registered margin 0.05; the gate-vs-residualization divergence on *identity_attack* is recorded, not adjudicated. Post-camera-ready, July 2026.

Implications for Secure AI deployment

- **Where to look:** attacks that repackage the same facts with different **salience** (euphemistic minimization).
- **Prompt-level defenses are bounded:** explicit warnings recover only $\sim 38\%$ — a ceiling co-occurring with universal overconfidence (ECE 0.19–0.42).
- **Ask the right question:** “*which* vulnerabilities matter for our use case?” — not “what’s the robustness score?”
- **Hardening target:** the fact-extraction front-end, since downstream kernels inherit its failures.

The larger program: *philosophy engineering*

What makes it engineering, not rhetoric

- A **space**, not a slogan: judgments are points, manipulation is **displacement**, robustness is **invariance**.
- **Pre-registration** + **admission gates that retract**: this talk's own bifactor test *failed* its *P1*; a channel rescue was *refuted*. Claims that do not transfer are **publicly demoted**.

The conjecture behind the program
(not tested here)

Normative and aesthetic structure is **geometric** — it lives in **compressibility** and **coarse-graining under an observer**.

One instrument, many domains: **harm** today; **aesthetics, law, cognition** on the same frame.

A philosophical claim you can **build, measure, and be proven wrong about** outranks one you can only defend.

Takeaways

Four takeaways

1. Vulnerabilities are **selective** — salience manipulation is the surface.
2. Robustness profiles are **dissociable** — and a single score is now *measured* to be mostly general valence.
3. The exploit is **end-to-end** — it flips scalar scores *and* typed verdicts.
4. It is **defensible** — class averaging halves natural-paraphrase drift; adversarial-at-scale is the registered open item.

Reproducible: `pip install agi-hpc (tag bds2026-v2) · moral-spectrum-analyzer · xbse · kernel: ErisML / DEME v3 · five tracks, ~8,000 calls under $50/day (Kaggle); validation panel on the open NRP managed LLM API.`



Thank you

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Questions?

Backup — grounding: 7-D harm space → DEME v3 9-D vector

Harm dim.	DEME v3 axis	Mapping
physical	physical harm	direct
autonomy	autonomy / consent	direct
trust	legitimacy / trust	direct
social	societal / environmental	direct
financial	fairness / equity	partial
emotional	virtue / care	partial
identity	privacy / standing	partial
—	rights, epistemic quality	not scored

The evaluation pipeline is the **measurement front-end**; the ErisML compiler's **DEME bridge** is the downstream decision kernel (DEMEVerdict).

Backup — the harm space is extensible *and* falsifiable

Discovery loop: residual analysis flags candidate missing dimensions → pre-registered admission gate.

- Scorecard: 3 flagged → **1 validated** (identity_attack), **1 retracted** (threat), **1 declined** (sexual content → policy channel); a registered rescue of the rights channel **refuted** (trained 0.509 vs null 0.512). **The gate really rejects — in both directions.**
- identity_attack: cross-dataset held-out AUROC **0.80** CI [0.78, 0.83] (+0.25 over null, 2 corpora, $n = 6400$) — wired live as a **10th channel**.
- Cross-lingual at scale: invariance index **0.72–0.80** across es/ar/zh/hi/sw, **harmful $\approx$ benign**.

Validated on the learned-encoder perception layer (xbse); native DEME re-run of the five tracks is the extended version. Post-camera-ready results, July 2026.